

AI AUTOMATION PORTFOLIO

Google Sheets & AI Agent Integration with n8n

1. About This Project

I created this project to show how we can use **Artificial Intelligence (AI)** to handle data automatically. Usually, people have to manually search through big Google Sheets to find information. My system solves this by using an **AI Agent** that can "talk" to the spreadsheet.

Main Goal: To build a smart assistant that can read user questions, search a database (Google Sheets), and give correct answers instantly.

2. How It Works (The Logic)

This automation follows a 4-step process to ensure everything works perfectly:

Step A: User Input

The process starts when a user sends a message in a chat box. For example: *"Check the status of invoice #123"*.

Step B: The AI Brain (Reasoning)

I used **Groq/OpenRouter** as the AI engine. It reads the user's message and decides: *"Do I need to check the Google Sheet for this?"*. If yes, it triggers the Google Sheets tool.

Step C: Database Action

The **n8n Google Sheets node** connects directly to my spreadsheet. It can find a specific row, update information, or add a new entry based on what the AI tells it to do.

Step D: Natural Response

Finally, the AI takes the data from the sheet and gives it back to the user in a friendly, human-like way instead of just showing raw data.

3. Key Features of the Workflow

- **Chat Memory:** I added a **Window Buffer Memory** node. This allows the AI to remember what the user said earlier in the conversation, making it feel like a real chat.
- **Tool Calling:** This is an advanced feature where the AI chooses the right tool (Google Sheets) only when it is needed.
- **Real-Time Updates:** Any change made in the Google Sheet is reflected in the chat answers immediately.

4. Technical Tools Used

- **n8n:** The automation platform used to connect everything.
- **Google Sheets:** Used as the primary database for storing information.
- **Groq / OpenRouter:** High-speed AI models for intelligent processing.
- **Chat Trigger:** To interact with the user via a simple chat interface.

5. What I Learned

By building this project, I improved my skills in:

- Connecting different apps using **APIs**.
- Designing logical workflows in **n8n**.

- Understanding how **LLMs (Large Language Models)** interact with external data.

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